

1. Project Overview and Goals

The primary objective of this project is to develop a data pipeline and predictive model to analyze and forecast domestic US commercial flight delays. By integrating historical performance data with weather observations, the system provides insights into the reasons behind air travel inefficiency

- Develop an end-to-end cloud-based data pipeline for large-scale ingestion and processing.
- Build a binary classifier to predict if a flight will arrive more than 15 minutes late.
- Construct an interactive dashboard for multi-dimensional analysis of delay causes and seasonal trends..

2. Dataset Description

The project uses two high-volume data sources:

- Airline On-Time Performance Data: Sourced from the US Bureau of Transportation Statistics, covering domestic flights from 2000 to the present.
- Airport Weather Data: Sourced from the NOAA to provide weather data such as visibility and wind speed.

Joined dataset:

☐	Field name	Type
☐	FlightDate	DATE
☐	Reporting_Airline	STRING
☐	Origin	STRING
☐	Dest	STRING
☐	CRSDepTime	INTEGER
☐	DepTime	FLOAT
☐	DepDelayMinutes	FLOAT
☐	CRSArrTime	INTEGER
☐	ArrTime	FLOAT
☐	ArrDelayMinutes	FLOAT
☐	Cancelled	INTEGER
☐	CarrierDelay	FLOAT
☐	WeatherDelay	FLOAT
☐	NASDelay	FLOAT
☐	SecurityDelay	FLOAT
☐	LateAircraftDelay	FLOAT
☐	AvgPrecip	FLOAT
☐	AvgVisibility	FLOAT
☐	AvgWindSpeed	FLOAT

Scale: The finalized dataset contains approximately 333 million rows and consumes roughly 70 GB of storage space, but I cleaned it down to 55 million to improve training speeds.

3. Pipeline Workflow and Services

stage	Service Used	Description
Ingestion	Google Cloud Storage	Initial location for raw CSV files.
Processing	Apache Spark	Used for parsing, joining 300+ monthly data files, and schema normalization.
Warehousing	BigQuery	Central repository partitioned by FlightDate for optimized analytical queries.
Machine Learning	Vertex AI	Classifier training and prediction using logistic regression.
Visualization	Looker Studio	Interactive dashboarding and exploratory data analysis.

4. Data Processing and Transformations

Raw data ingestion involved Python extraction scripts to download records from 2000-2025. I then converted the raw CSV data into Parquet format to optimize storage costs and improve query performance within BigQuery.

Key features for the pipeline include:

- Converting CRSDepTime integers into hour-of-day blocks to capture cyclical delay patterns.
- Weather Integration: Joining NOAA observations with BTS flight records based on airport codes and timestamps.
- Intermediate table used for type inference before joining aggregate table
- Removed rows with poor weather data to reduce row count from 333M to 55M

Resulting Table From Join + Cleaning:

FlightDate	Reporting_Airline	Origin	Dest	CRSDepTime	DepTime	DepDelayMinutes	CRSArrTime	ArrTime	ArrDelayMinutes	Cancelled	CarrierDelay	WeatherDelay
2006-08-18	YV	ORD	GSP	1815	1900.0	45.0	2112	2206.0	54.0	0	0.0	0
2015-11-17	DL	DEN	MSP	905	906.0	1.0	1200	1215.0	15.0	0	0.0	0
2007-09-13	DL	ATL	SAT	2125	2314.0	109.0	2250	50.0	120.0	0	0.0	0
2017-08-11	UA	ORD	LGA	1059	1106.0	7.0	1409	1417.0	8.0	0	null	n
2012-03-29	AA	ORD	SAT	1845	1844.0	0.0	2135	2134.0	0.0	0	null	n
2020-02-22	WN	LAX	DEN	910	910.0	0.0	1230	1218.0	0.0	0	null	n
2015-05-12	UA	ORD	IAD	830	834.0	4.0	1120	1107.0	0.0	0	null	n
2006-10-27	DL	ATL	SAV	925	1014.0	49.0	1026	1118.0	52.0	0	0.0	16
2009-09-21	MQ	ORD	CMH	1740	1738.0	0.0	1950	2006.0	16.0	0	0.0	0
2011-12-27	EV	ATL	SBN	1935	2035.0	60.0	2129	2223.0	54.0	0	0.0	0
2008-12-17	OO	ATL	PNS	730	755.0	25.0	749	837.0	48.0	0	0.0	0
2017-01-20	HA	LAX	HNL	745	815.0	30.0	1145	1207.0	22.0	0	0.0	0
2016-10-01	DL	ATL	AGS	2220	2218.0	0.0	2318	2303.0	0.0	0	null	n
2022-05-17	DL	ATL	ICT	1630	1627.0	0.0	1745	1729.0	0.0	0	null	n
2021-12-07	DL	ATL	BDL	1340	1342.0	2.0	1553	1540.0	0.0	0	null	n

Metrics: AVG of CarrierDelay, WeatherDelay, NASDelay, SecurityDelay, LateAircraftDelay.

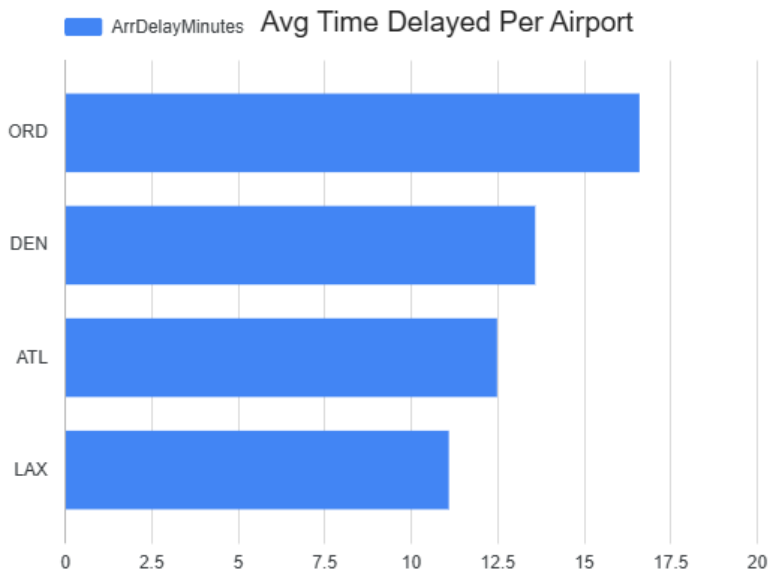


Chart: Airport Bottlenecks (Average Delay by Origin)

Metric: AVG(ArrDelayMinutes)

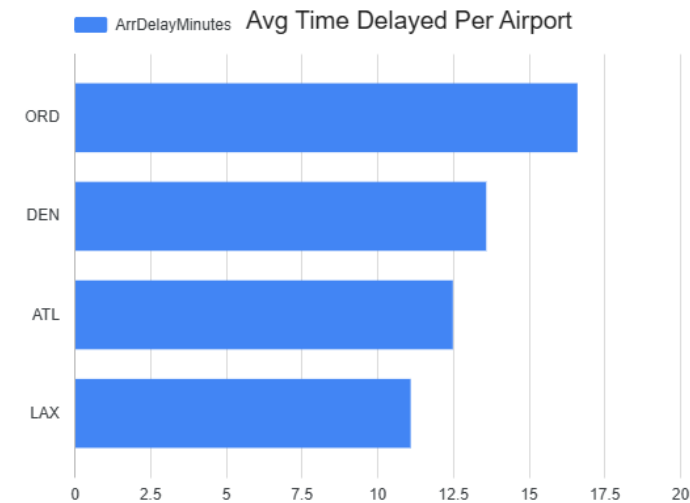


Chart: Delay Affects from Visibility (scatterplot)

Metrics: AvgVisibility, ArrDelayMinutes

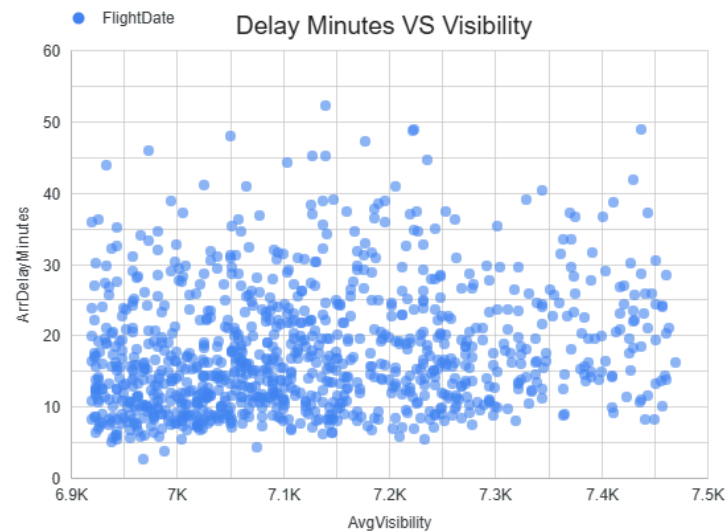
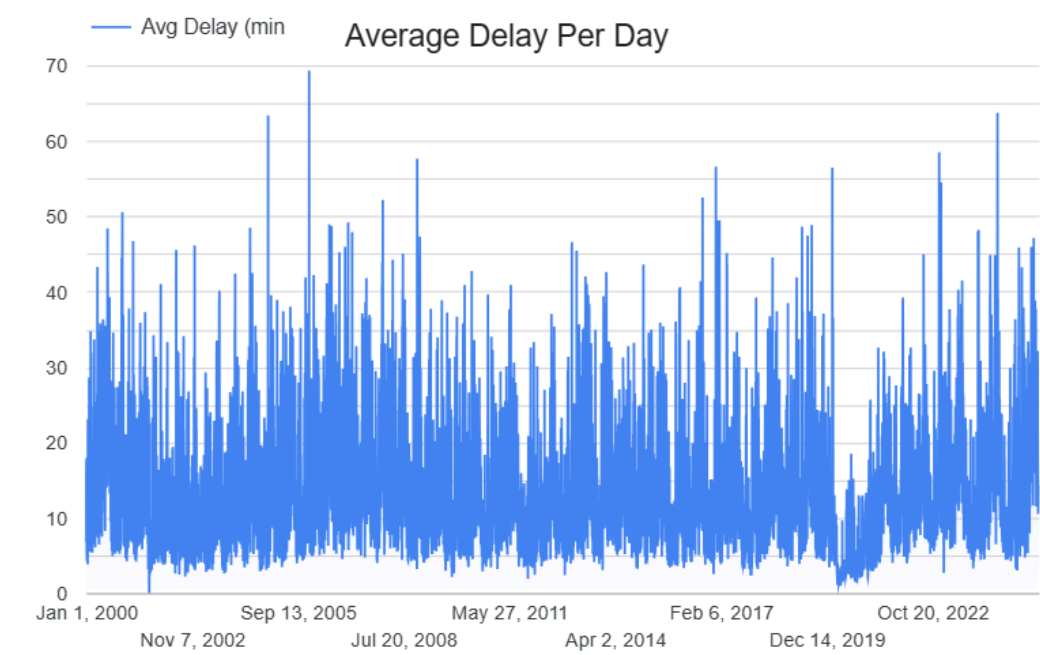


Chart: Avg Delay By Day (Time Series Chart)

Metrics: Flight Date, Avg Delay



Used a global filter for Cancelled = 0 so delay averages calculate actual flight performance rather than missing data points.

Logistic Regression Binary Classifier

Trained with a 80/20 train-test split using Reporting_Airline, Origin, Dest, CRSDepTime, AvgPrecip, AvgVisibility, and AvgWindSpeed to predict if a flight will be delayed or not

Label Column	
Filter Enter property name or value	
Field name	Type
predicted_is_delayed	INT64
Feature Columns	
Filter Enter property name or value	
Field name	Type
Reporting_Airline	STRING
Origin	STRING
Dest	STRING
CRSDepTime	INT64
AvgPrecip	FLOAT64
AvgVisibility	FLOAT64
AvgWindSpeed	FLOAT64

Evaluation:

Aggregate metrics 	
Threshold 	0.5000
Precision 	0.5000
Recall 	0.0149
Accuracy 	0.8121
F1 score 	0.0289
Log loss 	0.4576
ROC AUC 	0.6697

6. Challenges and Resolutions

Schema Mismatch: During the Spark-to-BigQuery loading phase, BigQuery initially rejected jobs due to Spark dynamically inferring data types incorrectly.

Resolution: The workflow was adjusted to load Parquet files into temporary tables, allowing BigQuery to auto-detect and finalize the schema before merging into the production table.

7. Lessons Learned & Next Steps

This project taught me the importance of schema management in big data pipelines.

In the future I would add:

- **Real-time Streams:** Integrating live flight feeds for real-time delay forecasting.
- **Feature Expansion:** Incorporating air traffic congestion metrics to improve classifier accuracy.